R108/A_prior_ce | TRAIN priors and actual latency

Complete forward including prior lookup and residual; no equal-gradient-scale or FLOPs claim.

Problem	Instances	Milliseconds / instance
TSP	128	0.249111
CVRP	128	0.300497
ATSP	128	0.068708
OVRP	128	0.068144
VRPB	128	0.068444
VRPL	128	0.068265
VRPTW	128	0.068089
OVRPTW	128	0.068267
OVRPB	128	0.068459
OVRPL	128	0.068136
VRPBL	128	0.068388
VRPBTW	128	0.068191
VRPLTW	128	0.068157
OVRPBL	128	0.068015
OVRPBTW	128	0.068165
OVRPLTW	128	0.068103
VRPBLTW	128	0.068319
OVRPBLTW	128	0.068183

Total trainable parameters: 1,842,953; pure cross entropy on s=logpi+t

Optional R94B: not_requested

TRAIN-only prior SHA256: a763633f8fd7b7177c7a5a1574c6e31dcd6e3ba25cb6940e1372081f45ea23de

R108 uses the original scratch R84/R88B full18 selector with 1,842,953 trainables. Both arms share the same complete fresh tensors, fixed TRAIN-only (problem, exactN) prior buffers, forward computation and original maximin inference on s=logpi+t. Active gradient freedom is not identical: CE common-shift gauge cancels the shared last-bias data gradient, whereas the ratio anchors the utility origin. t is the original learned utility. A trains pure CE(s); B trains the canonical product-noise binary ratio: half positive softplus(-t_y) plus half pi-weighted softplus(t). Its normalization differs from CE; no equal gradient-scale claim. The ratio sigmoid is not a champion probability. Training classification losses and evaluation maximin CE are different diagnostics and are not compared as equal losses. Single forward, dropout0.1; comparison/risk/KL/RDrop weights and RDrop ramp are zero. 57,600 updates, logical batch1024, seed2, AdamW1e-4 to1e-6. Prior tables are small frozen aggregate TRAIN-fit classifier parameters, not per-instance features or query labels. Priors alone do not demonstrate instance learning: the saved VAL/TEST t=0 reference is a TRAIN winner-frequency conditional prior, not a cost-optimal SBS. Model gains versus that reference are reported separately from gains versus SBS. No new feature cache, donor, additional path module, teacher, context labels, new targets, calibration, quota or auxiliary loss. R89A AdamW, R89B Muon and R84A are saved historical references, not rerun or newly parameter/FLOP-matched controls; R94B is optional, never a prerequisite. Fixed36 rescues, losses, false picks and aligned net costs/regret are separate from choice frequency. Native/FP64 ind and legal pools; TRAIN-fitted SBS; ALL task-equal. TEST is development-exposed; one whole-model full18 VAL checkpoint for full TRAIN/VAL/TEST/LIB; no TEST/LIB reselection. Actual standalone GPU-resident-input timing includes complete model prior lookup and residual computation, excludes dataset I/O, CPU transfer and solvers, and is not end-to-end dataset latency. No routing-summary requirement. No online W&B equality/availability is asserted.

R108/A_prior_ce | Same selected model: full TRAIN

Diagnostic only; TRAIN was never used to select the checkpoint; full TRAIN-fitted SBS.

Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
ALL	71.6171	66.0332	11.389125	11.400702	11.345073	-0.2006	+0.3842	0.3933
TSP	46.7716	32.8617	8.003733	8.023959	7.970215	-0.2521	+0.4205	0.4174
CVRP	48.8136	18.1284	18.042461	18.151987	17.946816	-0.6034	+0.5329	0.7044
ATSP	77.6247	46.9210	1.639216	1.678679	1.633938	-2.3508	+0.3230	0.3221
OVRP	41.6716	34.2420	7.443199	7.453531	7.397676	-0.1386	+0.6154	0.6267
VRPB	80.8049	80.2765	9.358941	9.357264	9.319342	+0.0179	+0.4249	0.4362
VRPL	38.6235	31.5309	12.391876	12.400882	12.344503	-0.0726	+0.3838	0.3904
VRPTW	80.5160	80.2407	18.812063	18.813703	18.757954	-0.0087	+0.2885	0.3024
OVRPTW	83.9543	83.8025	10.117825	10.118640	10.093220	-0.0081	+0.2438	0.2385
OVRPB	88.4617	88.3580	6.363010	6.363230	6.346658	-0.0035	+0.2576	0.2657
OVRPL	42.3617	34.4716	7.443279	7.453219	7.398483	-0.1334	+0.6055	0.6165
VRPBL	80.0654	79.4593	9.403301	9.403115	9.363196	+0.0020	+0.4283	0.4298
VRPBTW	79.7148	79.4914	19.731578	19.733549	19.642712	-0.0100	+0.4524	0.4364
VRPLTW	80.5679	80.3827	18.768153	18.769959	18.714366	-0.0096	+0.2874	0.3014
OVRPBL	88.6296	88.5556	6.358756	6.358857	6.342513	-0.0016	+0.2561	0.2645
OVRPBTW	83.3864	83.2432	10.623636	10.624446	10.586225	-0.0076	+0.3534	0.3278
OVRPLTW	83.8716	83.7321	10.110407	10.111265	10.085768	-0.0085	+0.2443	0.2415
VRPBLTW	79.8346	79.5630	19.768139	19.770814	19.680364	-0.0135	+0.4460	0.4285
OVRPBLTW	83.4346	83.3370	10.624679	10.625544	10.587365	-0.0081	+0.3524	0.3292

ALL task-equal; ratios of mean costs differ from per-instance actual regret.

VAL ALL vs TRAIN frequency prior (not SBS), positive = better: Top1 gain +2.1358 pp; cost gain +0.0129002; regret gain +0.1060 pp.

TEST ALL vs TRAIN frequency prior (not SBS), positive = better: Top1 gain +2.1728 pp; cost gain +0.0129303; regret gain +0.1092 pp.

R108/A_prior_ce | Fixed target36 vs R89A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1850	57	1998	0.000012450
VAL	4625	88	1	129	0.000738029
TEST	4598	92	6	143	0.000685364

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.372634	34889	29456	0.000012450	-0.001398
VAL	0.029630	1759	1735	0.000738029	0.003216
TEST	0.106173	1814	1728	0.000685364	0.006016

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R108/A_prior_ce | Fixed target36 vs R89B

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1504	404	1822	0.001048776
VAL	4625	72	22	115	0.001276009
TEST	4598	75	18	132	0.001000398

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.074005	28980	30059	0.001048776	0.009669
VAL	-0.167901	1531	1667	0.001276009	0.008221
TEST	-0.081481	1583	1649	0.001000398	0.010606

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R108/A_prior_ce | Fixed target36 vs R84A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1883	56	2018	-0.000035207
VAL	4625	92	1	131	0.000958815
TEST	4598	93	4	145	0.000627807

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.373251	34635	29193	-0.000035207	-0.001895
VAL	-0.004938	1737	1741	0.000958815	0.005403
TEST	0.102469	1793	1710	0.000627807	0.005405

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.